Example 10.27

Find the maximum value of the p.d.f of the Normal distribution with mean 40 and standard derivation 10. Also find its points of inflection.

Solution:

$\mu = 40 \qquad \sigma=10$

$$\text{Maximum value} = f(x) = \frac{1}{\sigma\sqrt{2\pi}} = \frac{1}{10\sqrt{2\pi}}$$

Points of inflection $= \mu \pm \sigma$

$= 40 \pm 10$

$= 30$ and 50

Example 10.28

A Normal variable X has the mean 50 and the S.D 5. Find its mean deviation about mean and the quartile deviation.

Solution:

Given Mean $=\mu=50$, standard deviation $\sigma = 5$

Mean deviation above mean $= 0.8\ \sigma$

$= 0.8 \times 5 = 4$

Quartile deviation $= 0.6745\sigma$

$= 0.6745 \times 5$

$= 3.3725$

Example 10.29

Find the quartiles of the Normal distribution having mean 60 and S.D 10.

Solution:

$\mu=60$, $\sigma=10$

Let x_1 be the value such that the area from x_1 to μ is 25%

$\therefore \qquad P(x_1 < X < \mu) = 25\% = 0.25$